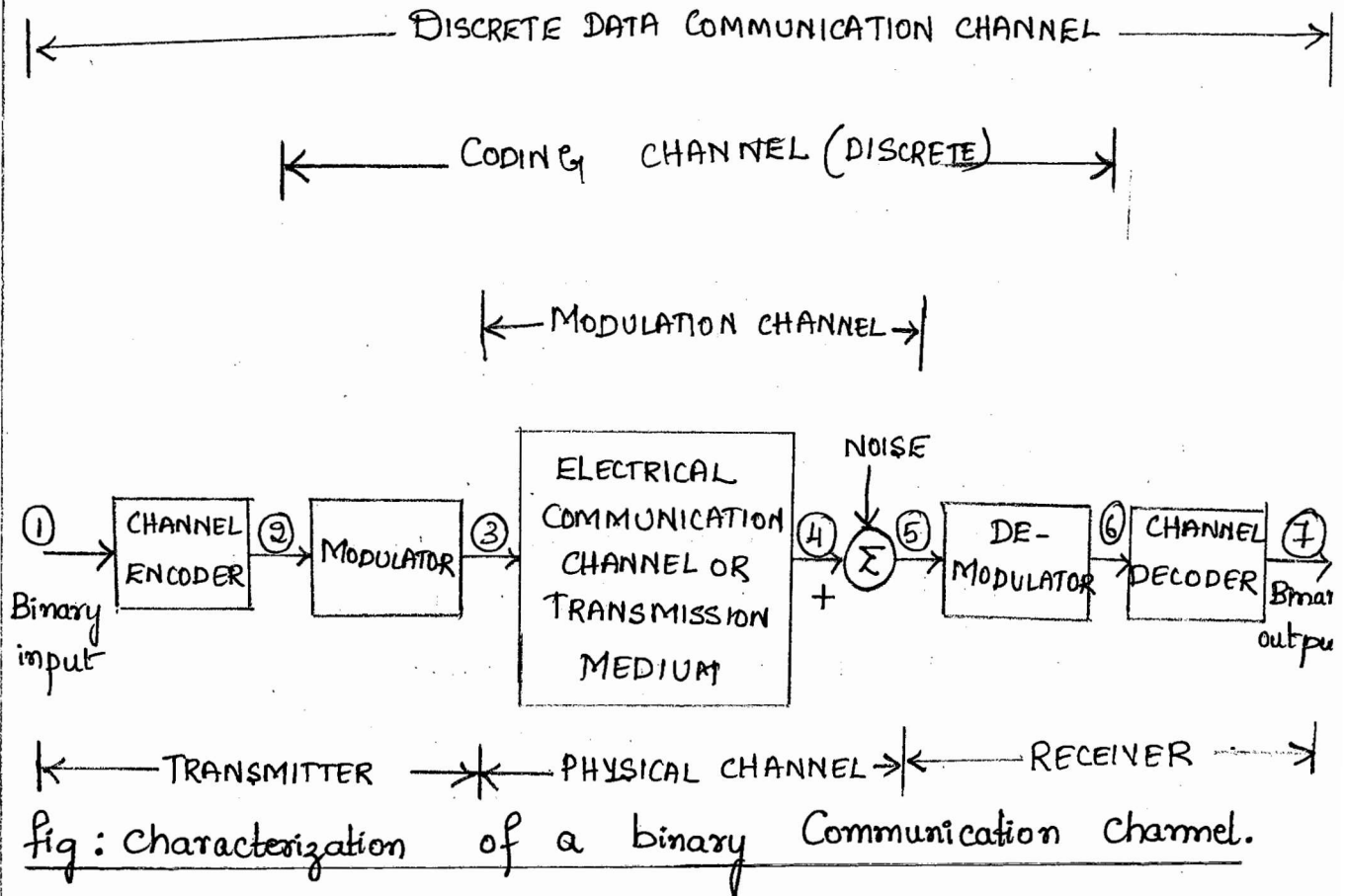


Fundamental limits on Performance & Channels for CommunicationCommunication Channels - IntroductionChethana.J
Asst Prof. RNSIT

- A practical Communication system can be divided into a Transmitter, physical channel and a Receiver.
- Transmitter consists of an encoder and a modulator, while the Receiver consists of demodulator and a decoder.
- B/w points 2 & 6 in the system shown there is a discrete often referred to as a coding channel that accepts a sequence of symbols at its input & produces a sequence of symbols at its output.
- The Communication channel b/w 3 & 5 in the system provides electrical connection b/w the transmitter and Receiver.
- The i/p and o/p are analog electrical waveforms. This portion of the channel is called a Continuous or modulation channel.
- The channel corrupts the signal statistically due to various types of additive & multiplicative noise.
- The important characteristic of a data communication system is the "channel capacity" which represents maximum rate @ which the data is transmitted across the channel with small probability error.

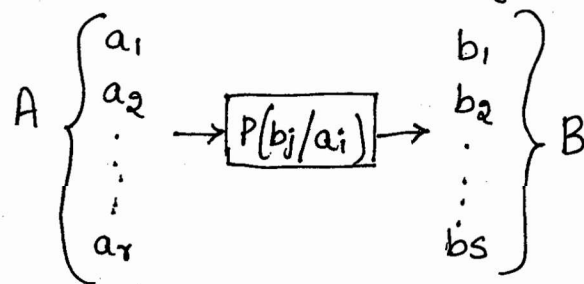
Discrete Communication Channels.

The input to the channel is a symbol belonging to an alphabet "A" with "n" symbols. The output of the symbol belonging to some other alphabet "B" with "s" symbols.

Due to error in the channel, the output symbol may be different from the input symbol. Errors are mainly due to noise in the analog portion of the channel.

Representation of a channel.

A Communication channel may be represented by a set of input alphabet $A = \{a_1, a_2, a_3, \dots, a_n\}$ consisting of 'n' symbols, a set of output alphabet $B = \{b_1, b_2, b_3, \dots, b_s\}$ consisting of 's' symbols and a set of conditional probabilities $P(b_j/a_i)$ with $i=1, 2, 3, \dots, n$ and $j=1, 2, \dots, s$ as shown below.

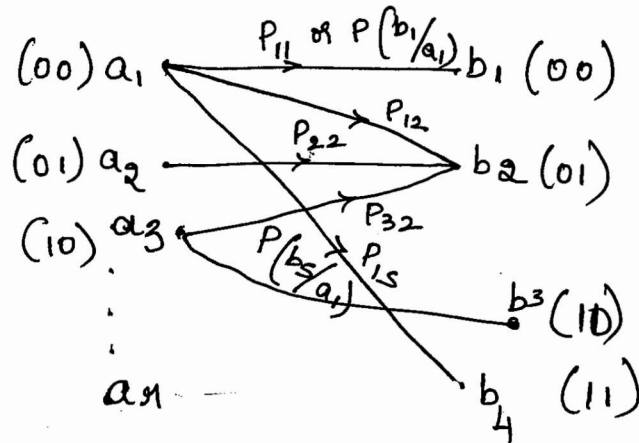


The Conditional probabilities existence is due to the presence of noise in the channel. We have a different number of symbols 's' at the receiver from 'n' symbols at the transmitter.

$\therefore$ we have, $n \times s$ number of Conditional probabilities which are represented in a "matrix" form with input symbols represented row-wise and output symbols column wise. Such a matrix is called "Channel Matrix or Noise Matrix".

$$P(b_j/a_i) \text{ or } P(B/A) = \begin{matrix} & \begin{matrix} b_1 & b_2 & \dots & b_s \end{matrix} \\ \begin{matrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{matrix} & \begin{bmatrix} P(b_1/a_1) & P(b_2/a_1) & \dots & P(b_s/a_1) \\ P(b_1/a_2) & P(b_2/a_2) & \dots & P(b_s/a_2) \\ \vdots & \vdots & \ddots & \vdots \\ P(b_1/a_n) & P(b_2/a_n) & \dots & P(b_s/a_n) \end{bmatrix} \end{matrix}$$

A channel can be represented with Conditional probabilities relating input and output symbols and the diagram. Showing these relation is known as channel/noise diagram.



In the above channel, the i/p symbol a_1 being transmitted having a code word "00". When these 2 binary symbols are transmitted over the channel, there is a chance that these 2 symbols are not affected and it is received as it is at the receiver.

this is the Conditional probability that symbol b_1 is received given that a_1 is transmitted at the transmitter denoted by $P(b_1/a_1)$. this can be written as

$$P_{11} = P(b_1/a_1) \rightarrow \textcircled{1}$$

but at the second transmission to a_2 there is a chance of 00 getting converted to 01. this can be written as

$$P_{12} = P(b_2/a_1) \rightarrow \textcircled{2}$$

$$P_{13} = P(b_3/a_1) \rightarrow \textcircled{3}$$

$$\& P_{14} = P(b_4/a_1) \rightarrow \textcircled{4}$$

But we know that ^{sum of all} probabilities = 1

$$P_{11} + P_{12} + P_{13} + P_{14} = 1 \rightarrow \textcircled{5}$$

using $\textcircled{1}$ to $\textcircled{4}$ in $\textcircled{5}$

$$P(b_1/a_1) + P(b_2/a_1) + P(b_3/a_1) + P(b_4/a_1) = 1 \rightarrow \textcircled{6}$$

Generalizing equation (6) for 's' output symbols and 'n' input symbols.

$$P(b_1/a_1) + P(b_2/a_1) + P(b_3/a_1) + \dots + P(b_s/a_1) = 1$$

writing with summation sign

$$\sum_{j=1}^s P(b_j/a_1) = 1$$

but it must hold good for all 'n' ^{i/p} symbols

$$\sum_{j=1}^s P(b_j/a_i) = 1 \quad \text{for } i = 1, 2, \dots, n$$

$\therefore$ the Sum of all the elements in any row of the channel Matrix is equal to unity.

Let us suppose take the probabilities of the i/p symbols namely $P(a_1), P(a_2), \dots, P(a_r)$ also $\sum_{i=1}^n P(a_i) = 1$. $\therefore P(a_1) + P(a_2) + \dots + P(a_r) = 1$

Knowing the channel matrix elements $P(b_j/a_i)$ for all i and j & the input probabilities $P(a_i)$ for all i, the probabilities of the output symbol $P(b_j)$ for all j, can be found by "theorem of total probability"

$$P(b_1) = P(b_1/a_1)P(a_1) + P(b_1/a_2)P(a_2) + \dots + P(b_1/a_n)P(a_n)$$

$$P(b_2) = P(b_2/a_1)P(a_1) + P(b_2/a_2)P(a_2) + \dots + P(b_2/a_n)P(a_n)$$

$$P(b_s) = P(b_s/a_1)P(a_1) + P(b_s/a_2)P(a_2) + \dots + P(b_s/a_r)P(a_r)$$

Knowing the input probabilities $P(a_i)$, Channel Matrix elements $P(b_j/a_i)$ and the output probabilities $P(b_j)$, the "Input Conditional Probabilities $P(a_i/b_j)$ " can be found,

By using "Baye's Rule."

$$P(a_i/b_j) = \frac{P(b_j/a_i)P(a_i)}{P(b_j)}$$

JOINT PROBABILITY.

The joint probability between any ^{input} symbol " a_i " and any output symbol " b_j " is given by

$$P(a_i, b_j) = P(b_j/a_i) P(a_i) = P(a_i/b_j) P(b_j) \rightarrow (1)$$

we know that

$$P(b_j/a_i) \text{ or } P(B/A) = \begin{matrix} & \begin{matrix} b_1 & b_2 & \dots & b_s \end{matrix} \\ \begin{matrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{matrix} & \begin{bmatrix} P(b_1/a_1) & P(b_2/a_1) & \dots & P(b_s/a_1) \\ P(b_1/a_2) & P(b_2/a_2) & \dots & P(b_s/a_2) \\ \vdots & \vdots & \ddots & \vdots \\ P(b_1/a_n) & P(b_2/a_n) & \dots & P(b_s/a_n) \end{bmatrix} \end{matrix} \rightarrow (2)$$

So multiplying, all the elements of 1st row of channel matrix by $P(a_1)$, 2nd row $P(a_2)$, ..., n^{th} row by $P(a_n)$ we get

$$P(b_j/a_i) P(a_i) = \begin{matrix} & \begin{matrix} b_1 & b_2 & \dots & b_s \end{matrix} \\ \begin{matrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{matrix} & \begin{bmatrix} P(b_1/a_1)P(a_1) & P(b_2/a_1)P(a_1) & \dots & P(b_s/a_1)P(a_1) \\ P(b_1/a_2)P(a_2) & P(b_2/a_2)P(a_2) & \dots & P(b_s/a_2)P(a_2) \\ \vdots & \vdots & \ddots & \vdots \\ P(b_1/a_n)P(a_n) & P(b_2/a_n)P(a_n) & \dots & P(b_s/a_n)P(a_n) \end{bmatrix} \end{matrix} \rightarrow (3)$$

using equation (1) in (3) we get

$$P(a_i, b_j) \text{ or } P(A, B) = \begin{matrix} & \begin{matrix} b_1 & b_2 & \dots & b_s \end{matrix} \\ \begin{matrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{matrix} & \begin{bmatrix} P(a_1, b_1) & P(a_1, b_2) & \dots & P(a_1, b_s) \\ P(a_2, b_1) & P(a_2, b_2) & \dots & P(a_2, b_s) \\ \vdots & \vdots & \ddots & \vdots \\ P(a_n, b_1) & P(a_n, b_2) & \dots & P(a_n, b_s) \end{bmatrix} \end{matrix} \rightarrow (4)$$

The elements in the matrix are the various joint probabilities between input and output symbols is called "JOINT PROBABILITY MATRIX" popularly denoted as JPM denoted by $P(a_i, b_j)$ or $P(A, B)$.

Properties of JPM.

① From the equation (3), we have

$$P(b_1) = P(b_1/a_1)P(a_1) + P(b_1/a_2)P(a_2) + \dots + P(b_1/a_r)P(a_r)$$

or from equn (4) we have

$$P(b_1) = P(a_1, b_1) + P(a_2, b_1) + \dots + P(a_r, b_1) \rightarrow (5)$$

The R.H.S of equn (5) are present in the 1st column of JPM. hence we can conclude that by adding all the elements of the 1st column, we get the probability of 1st output symbol b_1 .

$$\text{Similarly } P(b_2) = P(a_1, b_2) + P(a_2, b_2) + \dots + P(a_r, b_2) \rightarrow (6)$$

$$P(b_s) = P(a_1, b_s) + P(a_2, b_s) + \dots + P(a_r, b_s) \rightarrow (7)$$

from equn (5) to (7) we can derive the first property of JPM

i.e "BY adding the elements of JPM columnwise, we can obtain the probability of output symbols"

This property can be expressed as

$$\sum_{i=1}^r P(a_i, b_j) = P(b_j) \rightarrow (8)$$

for j varying from 1, 2, ..., S symbols

② Using the theorem of total probability for the input symbols, we have for the 1st symbol.

$$P(a_1) = P(a_1/b_1)P(b_1) + P(a_1/b_2)P(b_2) + \dots + P(a_1/b_s)P(b_s)$$

$\therefore$ from (1)

$$P(a_1) = P(a_1, b_1) + P(a_1, b_2) + \dots + P(a_1, b_s) \rightarrow (9)$$

$$\text{Similarly } P(a_2) = P(a_2, b_1) + P(a_2, b_2) + \dots + P(a_2, b_s) \rightarrow (10)$$

$$P(a_r) = P(a_r, b_1) + P(a_r, b_2) + \dots + P(a_r, b_s) \rightarrow (11)$$

∴ "By adding the elements of JPM row wise we can obtain the probability of input symbols"

Expressed as $\sum_{j=1}^S P(a_i, b_j) = P(a_i) \rightarrow (12)$
 for i varying from 1, 2, ..., r symbol

③ By adding eqn (9) to (11), we get

$$P(a_1) + P(a_2) + P(a_3) + \dots + P(a_r) = [P(a_1, b_1) + P(a_1, b_2) + \dots + P(a_1, b_s)] + [P(a_2, b_1) + P(a_2, b_2) + \dots + P(a_2, b_s)] + \dots + [P(a_r, b_1) + P(a_r, b_2) + \dots + P(a_r, b_s)] \quad (13) \leftarrow$$

where L.H.S of eqn (13) = 1

R.H.S is the sum of all the elements of JPM.

∴ "The sum of all the elements of JPM is equal to unity".

∴ Expressed as $\sum_{i=1}^r \sum_{j=1}^s P(a_i, b_j) = 1$

Problem.

1. In a Communication system, a transmitter has 3 input symbols $A = \{a_1, a_2, a_3\}$ and receiver also has 3 o/p symbols $B = \{b_1, b_2, b_3\}$. The matrix shows JPM with some marginal probabilities.

$a_i \backslash b_j$	b_1	b_2	b_3
a_1	$\frac{1}{12}$	*	$\frac{5}{36}$
a_2	$\frac{5}{36}$	$\frac{1}{9}$	$\frac{5}{36}$
a_3	*	$\frac{1}{6}$	*
$P(b_j)$	$\frac{1}{3}$	$\frac{14}{36}$	*

- (i) find the missing probabilities (*) in the table
 (ii) find $P(b_3/a_1)$ and $P(a_1/b_3)$
 (iii) Are the events a_1 and b_1 statistically independent? why?

Solⁿ: we know $\sum_{i=1}^3 P(a_i) = 1$ for i/p symbols
 similarly $\sum_{j=1}^3 P(b_j) = 1$ for o/p symbols.

(i) Given: $S = 3$
 $\therefore \sum_{j=1}^3 P(b_j) = P(b_1) + P(b_2) + P(b_3) = 1$
 $\therefore \frac{1}{3} + \frac{14}{36} + P(b_3) = 1$

$$P(b_3) = 5/18.$$

from first property of JPM we know, i.e. by adding the elements of JPM column wise. $\sum_{i=1}^3 P(a_i, b_j) = P(b_j)$
 $P(b_1) = P(a_1, b_1) + P(a_2, b_1) + P(a_3, b_1) \rightarrow 1^{st} \text{ column}$
 $\frac{1}{3} = \frac{1}{12} + \frac{5}{36} + P(a_3, b_1)$

$$P(a_3, b_1) = 1/9.$$

ii^{ly} $P(b_2) = P(a_1, b_2) + P(a_2, b_2) + P(a_3, b_2) \rightarrow 2^{nd} \text{ column}$
 $\frac{14}{36} = P(a_1, b_2) + \frac{1}{9} + \frac{1}{6}$

$$\therefore P(a_1, b_2) = 1/9.$$

iii^{ly} $P(b_3) = P(a_1, b_3) + P(a_2, b_3) + P(a_3, b_3) \rightarrow 3^{rd} \text{ column}$
 $\frac{5}{18} = \frac{5}{36} + \frac{5}{36} + P(a_3, b_3)$

$$\therefore P(a_3, b_3) = 0$$

	b_1	b_2	b_3
a_1	$1/12$	$1/9$	$5/36$
a_2	$5/36$	$1/9$	$5/36$
a_3	$1/9$	$1/6$	0
$P(b_i)$	$1/3$	$14/36$	$5/18$

(ii) To find $P(b_3/a_1)$ & $P(a_1/b_3)$

we have Conditional probability as,

$$P(b_j/a_i) = \frac{P(a_i, b_j)}{P(a_i)}$$

$$\therefore P(b_3/a_1) = \frac{P(a_1, b_3)}{P(a_1)}$$

from property (2) we have. By adding the elements of JPM Rowwise $\sum_{j=1}^3 P(a_i, b_j) = P(a_i)$
 $P(a_1) = P(a_1, b_1) + P(a_1, b_2) + P(a_1, b_3) \rightarrow 1^{st} \text{ row}$

$$P(a_1) = 1/12 + 1/9 + 5/36$$

$$P(a_1) = 1/3$$

$$\therefore P(b_3/a_1) = \frac{5/36}{1/3} \Rightarrow \boxed{\frac{5}{12}}$$

$$\text{III}^y \quad P(a_1/b_3) = \frac{P(a_1, b_3)}{P(b_3)} = \frac{(5/36)}{(5/18)} = \boxed{\frac{1}{2}}$$

(iii) To check whether a_1 & b_1 are statistically independent
 $P(a_1 \cap b_1) = P(a_1, b_1) = P(a_1) P(b_1)$

$$\text{consider } P(a_1) \cdot P(b_1) = (1/3)(1/3) = 1/9$$

$$\& \quad P(a_1, b_1) = 1/12$$

$$\therefore P(a_1, b_1) \neq P(a_1) P(b_1)$$

$\therefore a_1$ & b_1 are not statistically independent.

coz, we will receive b_1 by transmitting a_1 with some probability.

hence b_1 is dependent on a_1 .

Entropy Functions of Communication Channel.

- ① The input to the Communication Channel has 'n' symbols $(a_1, a_2, \dots, a_n)$ with probability of each symbol represented as $P(a_i)$ for i varying from 1 to n , then the average information or entropy of the input can be defined as

$$H(A) = \sum_{i=1}^n P(a_i) \log \left[\frac{1}{P(a_i)} \right] \text{ bits/symbol}$$

$$\& \quad 0 \leq H(A) \leq \log_2 n \quad \text{bits/symbol}$$

- ② The receiver receives 's' symbols $\{b_1, b_2, \dots, b_s\}$ and the probabilities are represented as $P(b_j)$ for j varying from 1 to s . The entropy of the receiver is defined as

$$H(B) = \sum_{j=1}^s P(b_j) \log \left[\frac{1}{P(b_j)} \right] \text{ bits/symbol}$$

$$\& \quad 0 \leq H(B) \leq \log_2 s \quad \text{bits/symbol}$$

- ③ Equivocation: The avg value of all the conditional entropies. Input and output Conditional entropies can be defined as

$$H(B/A) = \sum_{i=1}^n \sum_{j=1}^s P(a_i, b_j) \log \left(\frac{1}{P(b_j/a_i)} \right)$$

$$H(A/B) = \sum_{i=1}^n \sum_{j=1}^s P(a_i, b_j) \log \left(\frac{1}{P(a_i/b_j)} \right)$$

- ④ The joint entropy can be defined as

$$H(A, B) = \sum_{i=1}^n \sum_{j=1}^s P(a_i, b_j) \log \left(\frac{1}{P(a_i, b_j)} \right)$$

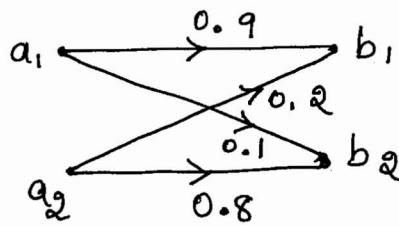
- ⑤ Relationships.

$$H(A, B) = H(B/A) + H(A)$$

$$H(A, B) = H(A/B) + H(B)$$

PROBLEMS

1. A binary channel is described by the following state diagram and given $P(a_1) = 0.7$ and $P(a_2) = 0.3$



find the entropies $H(A)$, $H(B)$, $H(A, B)$, $H(A/B)$, $H(B/A)$.

Solⁿ

$$P(B/A) = \begin{matrix} & b_1 & b_2 \\ \begin{matrix} a_1 \\ a_2 \end{matrix} & \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \end{matrix} \times \begin{matrix} 0.7 \\ 0.3 \end{matrix}$$

$\therefore$ "by multiplying the first row of the channel matrix with $P(a_1)$, by using theorem of total probability".

$$\therefore P(A, B) = \begin{bmatrix} 0.63 & 0.07 \\ 0.06 & 0.24 \end{bmatrix}$$

$$(i) H(A) = \sum_{i=1}^2 P(a_i) \log_2 \frac{1}{P(a_i)} = 0.7 \log_2 \frac{1}{0.7} + 0.3 \log_2 \frac{1}{0.3}$$

$$H(A) = 0.8813 \text{ bits/symbol}$$

$$(ii) H(B) = \sum_{j=1}^2 P(b_j) \log_2 \frac{1}{P(b_j)}$$

$$P(b_1) = 0.63 + 0.06 = 0.69 \quad P(b_2) = 0.07 + 0.24 = 0.31$$

$$\therefore H(B) = 0.69 \log_2 \frac{1}{0.69} + 0.31 \log_2 \frac{1}{0.31}$$

$$H(B) = 0.8931 \text{ bits/symbol}$$

$$(iii) H(A, B) = \sum_{i=1}^2 \sum_{j=1}^2 P(a_i, b_j) \log_2 \frac{1}{P(a_i, b_j)}$$

$$= 0.63 \log_2 \frac{1}{0.63} + 0.07 \log_2 \frac{1}{0.07} + 0.06 \log_2 \frac{1}{0.06} + 0.24 \log_2 \frac{1}{0.24}$$

$$H(A, B) = 1.426 \text{ bits/symbol}$$

$$\textcircled{iv} H(B/A)$$

we know

$$H(A, B) = H(B/A) + H(A)$$

$$H(B/A) = H(A, B) - H(A)$$

$$= 1.426 - 0.8813$$

$$H(B/A) = 0.5447 \text{ bits/symbol}$$

$$\textcircled{v} H(A/B)$$

we know

$$H(A, B) = H(A/B) + H(B)$$

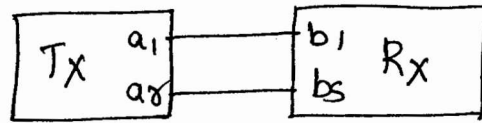
$$H(A/B) = H(A, B) - H(B)$$

$$= 1.426 - 0.8931$$

$$H(A/B) = 0.533 \text{ bits/symbol}$$

Mutual Information

Consider a transmitter sending 'n' symbols of data over a channel and the received symbols are 's' in number & may be different from those transmitted as shown,



Now the average information coming from the transmitter is given by $H(A)$

$$H(A) = \sum_{i=1}^r P(a_i) \log \frac{1}{P(a_i)} \text{ bits/symbol}$$

if the channel was noiseless then the average information received should be equal to $H(A)$.

When an average information of $H(A)$ is transmitted over the channel, an average amount of information equal to $H(A/B)$ is lost in the channel due to noise. ^{equivocation}

This factor is represented as the mutual information b/w the transmitter & the receiver & is defined as "Mutual Information" or "Transinformation".

$$I(A, B) = H(A) - H(A/B) \text{ bits/symbol.}$$

Properties of Mutual Information

① The mutual information of a channel is symmetric.

i.e. $I(A, B) = I(B, A)$

② The mutual information is always non-negative

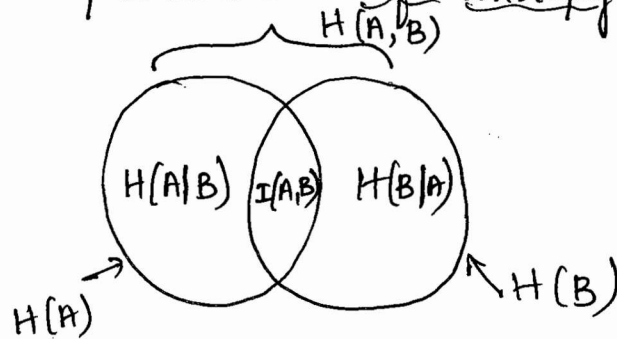
i.e. $I(A, B) \geq 0$

③ The mutual information of a channel may be expressed, for the entropy of the channel output as $I(A, B) = H(B) - H(B/A)$

- ④ The mutual information is related to the joint entropy of the channel,

$$I(A, B) = H(A) + H(B) - H(A, B)$$

Graphical representation of Entropy relations.



Problems.

1. A transmitter has an alphabet consisting of 5 letters $\{a_1, a_2, a_3, a_4, a_5\}$ and the receiver has an alphabet of four letters $\{b_1, b_2, b_3, b_4\}$. The joint probabilities of the systems are shown as.

$$P(A, B) = \begin{matrix} & b_1 & b_2 & b_3 & b_4 \\ \begin{matrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{matrix} & \begin{bmatrix} 0.25 & 0 & 0 & 0 \\ 0.10 & 0.30 & 0 & 0 \\ 0 & 0.05 & 0.10 & 0 \\ 0 & 0 & 0.05 & 0.10 \\ 0 & 0 & 0.05 & 0 \end{bmatrix} \end{matrix}$$

Compute different entropies of this channel.

→ Different Entropies are $H(A), H(B), H(A, B)$
 $H(B/A)$ & $H(A/B)$ &
 Mutual Information $I(A, B) = H(A) - H(A/B)$.

To find $H(A)$ first we have to find the probabilities of a_1 to a_5 , $\therefore$ from property-2 of JPM, we have.

$$\therefore P(a_1) = 0.25, P(a_2) = 0.40, P(a_3) = 0.15, P(a_4) = 0.15, P(a_5) = 0.05$$

Similarly $H(B)$, from property-1 of JPM

$$P(b_1) = 0.35, P(b_2) = 0.35, P(b_3) = 0.20, P(b_4) = 0.10$$

To find the entropy of i/p symbol

$$H(A) = \sum_{i=1}^5 P(a_i) \log_2 \frac{1}{P(a_i)} = 0.25 \log_2 \frac{1}{0.25} + 0.40 \log_2 \frac{1}{0.40} + 0.15 \log_2 \frac{1}{0.15} \\ + 0.15 \log_2 \frac{1}{0.15} + 0.05 \log_2 \frac{1}{0.05}$$

$$H(A) = 2.066 \text{ bits/symbol}$$

To find the entropy of o/p symbol

$$H(B) = \sum_{j=1}^4 P(b_j) \log_2 \frac{1}{P(b_j)} \\ = 0.35 \log_2 \frac{1}{0.35} + 0.35 \log_2 \frac{1}{0.35} + 0.20 \log_2 \frac{1}{0.20} + 0.10 \log_2 \frac{1}{0.10}$$

$$H(B) = 1.857 \text{ bits/symbol}$$

To find the joint entropy.

$$H(A, B) = \sum_{i=1}^5 \sum_{j=1}^4 P(a_i, b_j) \log_2 \frac{1}{P(a_i, b_j)}$$

$$H(A, B) = 0.25 \log_2 \frac{1}{0.25} + 0.10 \log_2 \frac{1}{0.10} + 0.30 \log_2 \frac{1}{0.30} \\ + 0.05 \log_2 \frac{1}{0.05} + 0.10 \log_2 \frac{1}{0.10} + 0.05 \log_2 \frac{1}{0.05} \\ + 0.05 \log_2 \frac{1}{0.05} + 0.10 \log_2 \frac{1}{0.10}$$

$$H(A, B) = 2.666 \text{ bits/symbol}$$

The equivocation $H(B/A)$ is

$$H(B/A) = H(A, B) - H(A) \\ = 2.666 - 2.066$$

$$H(B/A) = 0.6 \text{ bits/symbol}$$

The equivocation $H(A/B)$ is

$$H(A/B) = H(A, B) - H(B) \\ = 2.666 - 1.857$$

$$H(A/B) = 0.809 \text{ bits/symbol}$$

The mutual information $I(A, B)$ is found by

$$I(A, B) = H(A) - H(A/B) \\ = 2.066 - 0.809$$

$$I(A, B) = 1.257 \text{ bits/msg symbol}$$

(2) Show that $H(A, B) = H(A/B) + H(B)$

→ we know,

$$H(A/B) = \sum_{i=1}^r \sum_{j=1}^s P(a_i, b_j) \log \frac{1}{P(a_i/b_j)} \text{ bits/symbol} \rightarrow (1)$$

$$H(B) = \sum_{j=1}^s P(b_j) \log \frac{1}{P(b_j)} \text{ bits/symbol} \rightarrow (2)$$

$$\text{also } H(A, B) = \sum_{i=1}^r \sum_{j=1}^s P(a_i, b_j) \log \frac{1}{P(a_i, b_j)} \rightarrow (3)$$

$$\text{we know } P(a_i, b_j) = P(a_i/b_j) P(b_j) \rightarrow (4)$$

Replacing (4) in (3) we get

$$H(A, B) = \sum_{i=1}^r \sum_{j=1}^s P(a_i, b_j) \log \frac{1}{P(a_i/b_j) P(b_j)}$$

$$\therefore = \sum_{i=1}^r \sum_{j=1}^s P(a_i, b_j) \log \frac{1}{P(a_i/b_j)} + \sum_{i=1}^r \sum_{j=1}^s P(a_i, b_j) \log \frac{1}{P(b_j)}$$

from (1)

$$\Rightarrow H(A/B) + \sum_{j=1}^s \left[\sum_{i=1}^r P(a_i, b_j) \right] \log \frac{1}{P(b_j)}$$

↓
from JPM 1st property

$$\Rightarrow H(A/B) + \sum_{j=1}^s P(b_j) \log \frac{1}{P(b_j)}$$

$$\Rightarrow H(A/B) + H(B)$$

$$\therefore \boxed{H(A, B) = H(A/B) + H(B)}$$

- ③ A transmitter transmits five symbols with probabilities 0.2, 0.3, 0.2, 0.1 and 0.2. Given the channel matrix $P(B/A)$, calculate (i) $H(B)$ (ii) $H(A, B)$

$$P(B/A) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1/4 & 3/4 & 0 & 0 \\ 0 & 1/3 & 2/3 & 0 \\ 0 & 0 & 1/3 & 2/3 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

→ from JPM eqn we have

$$P(a_i, b_j) = P(a_i) P(b_j/a_i) \rightarrow \textcircled{1}$$

∴ multiplying 1st row of $P(B/A)$ with 0.2,
2nd row with 0.3
3rd row " 0.1
4th row " 0.2

the JPM $P(a_i, b_j)$ can be constructed.

$$P(B/A) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1/4 & 3/4 & 0 & 0 \\ 0 & 1/3 & 2/3 & 0 \\ 0 & 0 & 1/3 & 2/3 \\ 0 & 0 & 1 & 0 \end{bmatrix} \times \begin{matrix} 0.2 \\ 0.3 \\ 0.2 \\ 0.1 \\ 0.2 \end{matrix} \Rightarrow P(a_i, b_j) = \begin{bmatrix} 1/5 & 0 & 0 & 0 \\ 3/40 & 9/40 & 0 & 0 \\ 0 & 1/15 & 2/15 & 0 \\ 0 & 0 & 1/30 & 1/15 \\ 0 & 0 & 1/5 & 0 \end{bmatrix}$$

- ① To find $H(B)$, Adding the elements of each column, we get
 $P(b_1) = 1/5 + 3/40 = 11/40$, $P(b_2) = 7/24$, $P(b_3) = 11/30$, $P(b_4) = 1/15$

$$\therefore H(B) = \sum_{j=1}^4 P(b_j) \log_2 \frac{1}{P(b_j)} \Rightarrow \frac{11}{40} \log_2 \frac{40}{11} + \frac{7}{24} \log_2 \frac{24}{7} + \frac{11}{30} \log_2 \frac{30}{11} + \frac{1}{15} \log_2 15$$

$$\therefore H(B) = 1.822 \text{ bits/symbol}$$

- ② To find $H(A, B) \Rightarrow H(A, B) = \sum_{i=1}^5 \sum_{j=1}^4 P(a_i, b_j) \log \frac{1}{P(a_i, b_j)}$

$$= \frac{1}{5} \log_2 5 + \frac{3}{40} \log_2 \frac{40}{3} + \frac{9}{40} \log_2 \frac{40}{9} + \frac{1}{15} \log_2 15 + \frac{2}{15} \log_2 \frac{15}{2} + \frac{1}{3} \log_2 30 + \frac{1}{5} \log_2 5 + \frac{1}{15} \log_2 15$$

$$\therefore H(A, B) = 2.7653 \text{ bits/sym}$$

- ④ For the JPM given, compute individually $H(A)$, $H(B)$, $H(A, B)$, $H(A/B)$, $H(B/A)$ and $I(A, B)$. Verify the relationship among these entropies.

$$P(A, B) = \begin{bmatrix} 0.05 & 0 & 0.20 & 0.05 \\ 0 & 0.10 & 0.10 & 0 \\ 0 & 0 & 0.20 & 0.10 \\ 0.05 & 0.05 & 0 & 0.10 \end{bmatrix}$$

→ From property - 1, adding columnwise

$$P(b_1) = 0.10, P(b_2) = 0.15, P(b_3) = 0.50, P(b_4) = 0.25$$

$$\therefore H(B) = \sum_{j=1}^4 P(b_j) \log_2 \frac{1}{P(b_j)} \quad (\text{The entropy of } p \text{ symbol})$$

$$= 0.10 \log_2 \frac{1}{0.10} + 0.15 \log_2 \frac{1}{0.15} + 0.50 \log_2 \frac{1}{0.50} + 0.25 \log_2 \frac{1}{0.25}$$

$$\boxed{H(B) = 1.743 \text{ bits/symbol}}$$

From property - 2, adding rowwise.

$$P(a_1) = 0.30, P(a_2) = 0.20, P(a_3) = 0.30, P(a_4) = 0.20$$

$$\therefore H(A) = \sum_{i=1}^4 P(a_i) \log_2 \frac{1}{P(a_i)} \quad (\text{The entropy of } i/p \text{ symbol})$$

$$= 0.30 \log_2 \frac{1}{0.30} + 0.20 \log_2 \frac{1}{0.20} + 0.30 \log_2 \frac{1}{0.30} + 0.20 \log_2 \frac{1}{0.20}$$

$$\boxed{H(A) = 1.971 \text{ bits/symbol}}$$

The Joint entropy is given by

$$H(A, B) = \sum_{i=1}^4 \sum_{j=1}^4 P(a_i, b_j) \log_2 \frac{1}{P(a_i, b_j)}$$

$$= \left(0.05 \log_2 \frac{1}{0.05}\right) \times 4 + \left(0.10 \log_2 \frac{1}{0.10}\right) \times 4 + \left(0.2 \log_2 \frac{1}{0.2}\right) \times 2$$

$$\boxed{H(A, B) = 3.122 \text{ bits/symbol}}$$

$$H(A/B) = H(A, B) - H(B)$$

$$= 3.122 - 1.743$$

$$H(A/B) = 1.379 \text{ bits/symbol}$$

$$H(B/A) = H(B, A) - H(A)$$

$$= 3.122 - 1.971$$

$$H(B/A) = 1.151 \text{ bits/symbol}$$

we can also calculate $H(A/B)$ & $H(B/A)$ by using equivocation

$$\therefore \text{we have } H(A/B) = \sum_{i=1}^4 \sum_{j=1}^4 P(a_i, b_j) \log_2 \frac{1}{P(a_i, b_j)}$$

$$\therefore \text{using relationship } P(a_i | b_j) = \frac{P(a_i, b_j)}{P(b_j)}$$

the $P(A/B)$ is constructed.

$$P(A/B) = \begin{bmatrix} \frac{0.05}{0.10} & 0 & \frac{0.2}{0.5} & \frac{0.05}{0.25} \\ 0 & \frac{0.10}{0.15} & \frac{0.1}{0.5} & 0 \\ 0 & 0 & \frac{0.2}{0.5} & \frac{0.10}{0.25} \\ \frac{0.05}{0.10} & \frac{0.05}{0.15} & 0 & \frac{0.10}{0.25} \end{bmatrix} \Rightarrow \begin{bmatrix} 1/2 & 0 & 2/5 & 1/5 \\ 0 & 2/3 & 1/5 & 0 \\ 0 & 0 & 2/5 & 2/5 \\ 1/2 & 1/3 & 0 & 2/5 \end{bmatrix}$$

$$\therefore H(A/B) = \sum_{i=1}^4 \sum_{j=1}^4 P(a_i, b_j) \log_2 \frac{1}{P(a_i | b_j)}$$

$$\Rightarrow H(A/B) = 0.05 \log_2 2 + 0.05 \log_2 2 + 0.10 \log_2 \frac{3}{2} + 0.05 \log_2 3$$

$$+ 0.20 \log_2 \frac{5}{2} + 0.10 \log_2 5 + 0.20 \log_2 \frac{5}{2} + 0.05 \log_2 5$$

$$+ 0.10 \log_2 \frac{5}{2} + 0.10 \log_2 \frac{5}{2}$$

$$H(A/B) = 1.379 \text{ bits/message symbol}$$

Similarly $P(B/A) = \frac{P(a_i, b_j)}{P(A)}$

$$\Rightarrow P(B/A) = \begin{bmatrix} \frac{0.05}{0.30} & 0 & \frac{0.20}{0.30} & \frac{0.05}{0.30} \\ 0 & \frac{0.10}{0.20} & \frac{0.10}{0.20} & 0 \\ 0 & 0 & \frac{0.20}{0.30} & \frac{0.10}{0.30} \\ \frac{0.05}{0.20} & \frac{0.05}{0.20} & 0 & \frac{0.10}{0.20} \end{bmatrix} = \begin{bmatrix} 1/6 & 0 & 2/3 & 1/6 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 2/3 & 1/3 \\ 1/4 & 1/4 & 0 & 1/2 \end{bmatrix}$$

$$\therefore P(B/A) = 1.151 \text{ bits/message symbol.}$$

$\therefore$ The mutual Information $I(A, B)$ is given by

$$I(A, B) = H(A) - H(A/B)$$

$$= 1.971 - 1.379$$

$$I(A, B) = 0.592 \text{ bits/message symbol}$$

Rate of Information Transmission over a Discrete Channel

If a discrete memoryless channel accepts symbols at the rate of ' η_s ' symbols/sec, then the average rate at which information is going into the channel is given by

$$R_{in} = H(A) \eta_s \text{ bits/second (bps)}$$

But at the receiver, since it is not possible to reconstruct the symbol sequence with certainty the error introduced by the channel also known as "Equivocation" represented by $H(A/B)$ should be taken into account. hence net information received in the mutual information, given as:

$$I(A, B) = H(A) - H(A/B) \text{ bits/message-symbol}$$

hence the ^{average rate of} information transmission " R_t " is given

$$\text{by } R_t = I(A, B) \eta_s \text{ bits/second.}$$

* Channel Capacity / Capacity of a Discrete Memoryless channel

- The capacity of a discrete memoryless noisy channel is defined as the maximum possible rate of information transmission over the channel.

The maximum rate of transmission occurs when the source is "matched" to the channel.

∴ Channel Capacity C is defined as

$$C = \text{Max} [R_t] \text{ bps}$$

i.e. $C = \text{Max} [I(A, B)] \eta_s \text{ bps}$

$$C = \text{Max} [H(A) - H(A/B)] \eta_s \text{ bits per second.}$$

* Shannon's theorem on Channel Capacity [Shannon's Second theorem]

It is stated in 2 ways

- Positive Statement: when the rate of information transmission is less than or equal to the capacity of the channel, then there exists a coding technique which enables transmission over the channel with as small a probability of error as possible even in the presence of noise $R \leq C$

- Negative Statement: If $R > C$, then reliable transmission of information is not possible and no coding technique can control the errors.

* Channel Efficiency

The "channel efficiency" denoted by η_{ch} is given by

$$\eta_{ch} = \frac{R_t}{C} \times 100\%$$

Substituting for R_t and C

we have

$$\eta_{ch} = \frac{[H(A) - H(A/B)] \eta_s}{\text{Max}[H(A) - H(A/B)] \eta_s} \times 100\%$$

or

$$\eta_{ch} = \frac{I(A, B)}{\text{Max}[I(A, B)]} \times 100\%$$

* Channel Redundancy

The channel Redundancy denoted by R_{ch} is given by

$$R_{ch} = 1 - \eta_{ch} \quad \text{where } \eta_{ch} \text{ is a fraction.}$$

Special Channels

There are several special channels which are in the field of communication systems. Some of them are listed below here we need to find the channel capacity of these channels

- 1) Symmetric/uniform channels
- 2) Binary Symmetric channels (BSC)
- 3) Binary Erasure channels (BEC)
- 4) Noiseless channel
- 5) Deterministic channel
- 6) Cascaded channel.

Capacity of different types of Channels.

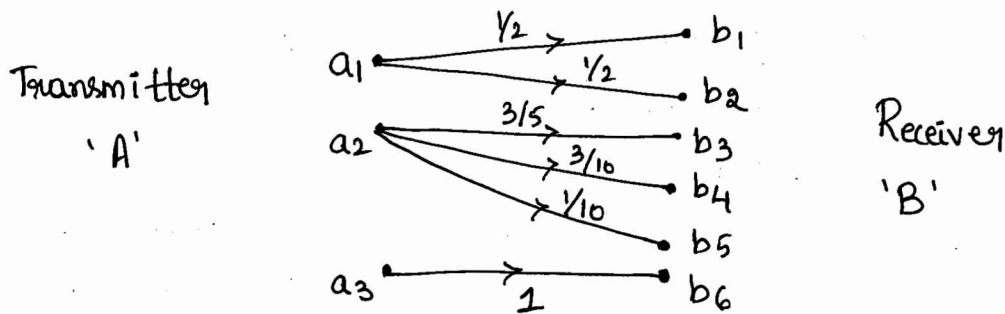
1) Noiseless Channel: This channel is represented by a channel Matrix with one & only one non-zero element "every column" is defined as "noiseless channel".

It has 2 properties.

- ① Each column contains only one element with other elements in that column being zero
- ② the sum of all the elements in any row is equal to unity.

ex:- $P(B/A)$ or $P(b_j/a_i) =$

a_1	b_1	b_2	b_3	b_4	b_5	b_6
	$1/2$	$1/2$	0	0	0	0
a_2	0	0	$3/5$	$3/10$	$1/10$	0
a_3	0	0	0	0	0	1



Since it is a noiseless channel, the amount of information lost in the channel due to noise is zero, hence the channel becomes a noiseless channel.

$\therefore$ the Mutual information $I(A,B) = H(A) - H(A/B)^0$
 $I(A,B) = H(A)$

$\therefore$ Channel capacity $C = \text{Max } (I(A,B)) \text{ n/s}$

$C = (\text{Max } H(A)) \text{ n/s}$. $H(A)_{\text{max}} = \log_2 8$

considering '8' i/p symbols $C = \log_2 8 \cdot 1 \text{ n/s}$ let $1 \text{ n/s} = 1 \text{ sym/sec}$

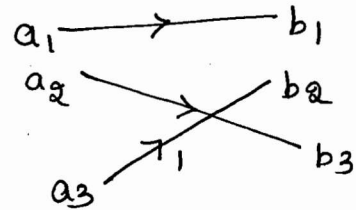
$C = \log_2 8 \text{ bits/sec}$

2) Deterministic Channel: A channel represented by a channel matrix with one & only one non zero element in "every row" is defined as "Deterministic Channel".

A deterministic channel must satisfy these 2 properties.

- (1) The sum of the elements in every row must be equal to 1 &
- (2) Each row must contain only one entry & the other elements in that row must be zero.

ex:- $P(B/A) = \begin{matrix} & b_1 & b_2 & b_3 \\ \begin{matrix} a_1 \\ a_2 \\ a_3 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \end{matrix}$



In the deterministic channel, one can "determine" which symbol is to be received at the receiver & hence the channel is called deterministic channel. here the channel is characterized by the conditional entropy $H(B/A) = 0$.

$\therefore$ the mutual information of a deterministic channel is given by

$$I(A, B) = H(B) - H(B/A)$$

$$\therefore I(A, B) = H(B)$$

the channel capacity C is given by

$$C = \text{Max}[I(A, B)]_{\eta_s}$$

$$C = \text{Max}[H(B)]_{\eta_s}$$

$$\therefore C = \log_2 S \cdot \eta_s$$

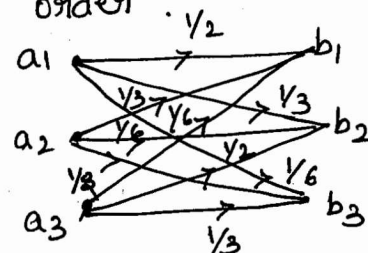
if $\eta_s = 1 \text{ sym/sec}$

$$C = \log_2 S \text{ bits/symbol.}$$

$\therefore H(B) \text{ max o/p}$
symbol will be 'S'

- (3) Symmetric/Uniform Channel : A channel is said to be symmetric or uniform channel, if the second & subsequent rows of the channel matrix contains the same elements as that of first row, but in a different order.

ex:- $P(B/A) = \begin{matrix} & b_1 & b_2 & b_3 \\ \begin{matrix} a_1 \\ a_2 \\ a_3 \end{matrix} & \begin{bmatrix} 1/2 & 1/3 & 1/6 \\ 1/3 & 1/6 & 1/2 \\ 1/6 & 1/2 & 1/3 \end{bmatrix} \end{matrix}$



Channel Capacity of symmetric/Uniform channel

In general the channel matrix of a symmetric/uniform channel can be written as .., for n i/p symbols & S o/p symbols.

$$P(B/A) = \begin{matrix} & \begin{matrix} b_1 & b_2 & b_3 & \dots & b_S \end{matrix} \\ \begin{matrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{matrix} & \begin{bmatrix} P_1 & P_2 & P_3 & \dots & P_S \\ P_2 & P_3 & P_{S-1} & \dots & P_1 \\ P_4 & P_{S-1} & P_{S-4} & \dots & P_{S-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ P_3 & P_{S-1} & P_{S-2} & \dots & P_1 \end{bmatrix} \end{matrix} \rightarrow (1)$$

where $P_1, P_2, P_3, \dots, P_{S-4}, P_{S-3}, P_{S-2}, P_{S-1}$ are the conditional probabilities of $P(b_j/a_i)$,

we know the sum of all the elements of any row equal to unity

$$\sum_{j=1}^S P(b_j/a_i) = 1 \rightarrow (2)$$

But all the ' n ' number of rows have same elements as in the first row, it can be written as

$$\sum_{j=1}^S P_j = 1 \rightarrow (3)$$

we also know $P(a_i, b_j) = P(b_j/a_i) P(a_i)$ from Joint Probability

$\therefore$ The equivocation $H(B/A)$ is given by

$$H(B/A) = \sum_{i=1}^n \sum_{j=1}^S P(a_i, b_j) \log_2 \frac{1}{P(b_j/a_i)} \rightarrow (4)$$

Applying (A) in (4)

$$H(B/A) = \sum_{i=1}^n \sum_{j=1}^S P(b_j/a_i) P(a_i) \log_2 \frac{1}{P(b_j/a_i)}$$

$$= \underbrace{\sum_{i=1}^n P(a_i)}_{\rightarrow 1} \sum_{j=1}^S \underbrace{P(b_j/a_i)}_{\text{from (3)}} \log_2 \frac{1}{P(b_j/a_i)}$$

$$H(B/A) = \sum_{j=1}^S P_j \log_2 \frac{1}{P_j}$$

$$\Rightarrow \boxed{H(B/A) = h}$$

$$\text{where } h = \sum_{j=1}^S P_j \log_2 \frac{1}{P_j}$$

∴ The mutual information $I(A,B)$ is given by

$$I(A,B) = H(B) - H(B|A)$$

$$I(A,B) = H(B) - h$$

The channel capacity $C = \text{Max} [I(A,B)] \eta_s$

$$C = \text{Max} [H(B) - H(B|A)] \eta_s$$

$$C = \text{Max} (H(B) - h) \eta_s \quad \text{if } \eta_s = 1 \text{ sy/sec}$$

$$C = \text{Max} H(B) - h \quad \text{since } h \text{ is constant}$$

∴ we know from Deterministic channel $H_B \text{Max} = \log_2 S$

$$\therefore H(B)_{\text{max}} = \log_2 S$$

$$\therefore \boxed{C = \log_2 S - h \text{ bits/sec}} \quad \text{for } h = \sum_{j=1}^S P_j \log_2 \frac{1}{P_j}$$

Problems

① For the channel Matrix shown, find the channel Capacity

$$P(b_j/a_i) = \begin{matrix} & \begin{matrix} b_1 & b_2 & b_3 \end{matrix} \\ \begin{matrix} a_1 \\ a_2 \\ a_3 \end{matrix} & \begin{bmatrix} 1/2 & 1/3 & 1/6 \\ 1/3 & 1/6 & 1/2 \\ 1/6 & 1/2 & 1/3 \end{bmatrix} \end{matrix}$$

→ The given matrix belongs to a symmetric or uniform channel.

$$H(B|A) = h = \sum_{j=1}^S P_j \log_2 \frac{1}{P_j}$$

$$= \frac{1}{2} \log_2 2 + \frac{1}{3} \log_2 3 + \frac{1}{6} \log_2 6$$

$$h = 1.4591 \text{ bits/symbol}$$

The channel capacity C is given by

$$C = (\log_2 S - h) \eta_s$$

$$C = (\log_2 3 - 1.4591) \eta_s$$

$$\boxed{C = 0.1258 \text{ bits/sec}} \quad \text{with } \eta_s = 1 \text{ sy/sec}$$

- ② For the channel matrix given below, compute the channel capacity.

$$P(Y/X) = \begin{bmatrix} 0.6 & 0.2 & 0.2 \\ 0.2 & 0.6 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix} \text{ with } \eta_S = 1000 \text{ symbols/sec.}$$

→ The given matrix is that of a symmetric channel

$$\therefore h = \sum_{j=1}^3 P_j \log \frac{1}{P_j}$$

$$= 0.6 \log \frac{1}{0.6} + 2 \times 0.2 \log \frac{1}{0.2}$$

$$h = 1.371 \text{ bits/message-symbol}$$

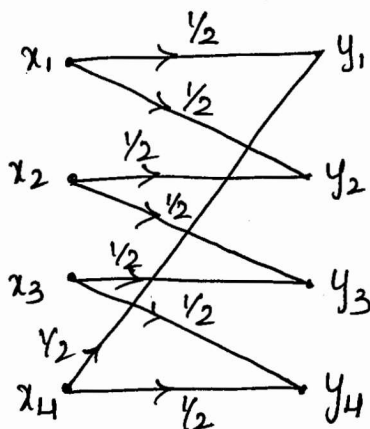
Given S = number of output symbols = 3

$$\therefore \text{Channel Capacity } C = (\log S - h) \eta_S \text{ bits/sec}$$

$$= (\log 3 - 1.371)(1000) \text{ bits/sec}$$

$$\therefore C = 214 \text{ bits/sec}$$

- ③ Determine the capacity of the channel shown



→ Solution ∴ The channel matrix $P(Y/X)$ can be written as

$$P(Y/X) = \begin{matrix} & \begin{matrix} y_1 & y_2 & y_3 & y_4 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{matrix} & \begin{bmatrix} 1/2 & 1/2 & 0 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 1/2 & 0 & 0 & 1/2 \end{bmatrix} \end{matrix}$$

The 2nd, 3rd and 4th rows of the matrix consists of the same elements as the 1st row elements.

hence the given channel is a symmetric/uniform channel.

$$h = H(Y/X) = \sum_{j=1}^4 P_j \log \frac{1}{P_j}$$

$$h = \frac{1}{2} \log 2 + \frac{1}{2} \log 2 + 0 + 0$$

$$\boxed{h = 1 \text{ bit/message-symbol}}$$

∴ The channel Capacity is given by

$$C = \log_2 S - h$$

$$= \log_2 4 - 1$$

$$= 2 - 1$$

$$\boxed{C = 1 \text{ bit/message-symbol}}$$

④ A channel has the following characteristics.

$$P(Y/X) = \begin{matrix} & Y_1 & Y_2 & Y_3 & Y_4 \\ \begin{matrix} X_1 \\ X_2 \end{matrix} & \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} \end{matrix}$$

find $H(X)$, $H(Y)$, $H(X, Y)$ and channel capacity if

$\eta = 1000$ symbols/sec.

To get $P(X, Y) = \begin{matrix} & Y_1 & Y_2 & Y_3 & Y_4 \\ \begin{matrix} X_1 \\ X_2 \end{matrix} & \begin{bmatrix} \frac{1}{6} & \frac{1}{6} & \frac{1}{12} & \frac{1}{12} \\ \frac{1}{12} & \frac{1}{12} & \frac{1}{6} & \frac{1}{6} \end{bmatrix} \end{matrix}$ $P(X_1) = \frac{1}{2}$
 $P(X_2) = \frac{1}{2}$

The o/p probabilities are all equal given by

$$P(Y_1) = P(Y_2) = P(Y_3) = P(Y_4) = \frac{1}{6} + \frac{1}{12} = \frac{1}{4}$$

$$\therefore H(X) = \sum_{i=1}^2 P(X_i) \log \frac{1}{P(X_i)} \Rightarrow \frac{1}{2} \log 2 + \frac{1}{2} \log 2$$

$$\boxed{H(X) = 1 \text{ bit/message-symbol}}$$

$$H(Y) = \sum_{j=1}^4 P(Y_j) \log \frac{1}{P(Y_j)} \Rightarrow \frac{1}{4} \log 4 \times 4 \Rightarrow 2 \text{ bits/message symbol}$$

$$\boxed{H(Y) = 2 \text{ bits/message symbol}}$$

∴ The joint entropy $H(X, Y)$ is given by eqnⁿ as

$$H(X, Y) = \sum_{i=1}^2 \sum_{j=1}^4 P(x_i, y_j) \log \frac{1}{P(x_i, y_j)}$$

$$H(X, Y) = \left(\frac{1}{6} \log 6 \right) (4) + \left(\frac{1}{12} \log 12 \right) (4)$$

$$H(X, Y) = 2.9183 \text{ bits/message-symbol}$$

The mutual information is given by

$$I(X, Y) = H(X) + H(Y) - H(X, Y)$$

The channel capacity given by

$$C = \text{Max} [I(X, Y)] = 1 + 2 - 2.9183$$

$$C = 0.0817 \text{ bits/message-symbol}$$

Given $\eta = 1000 \text{ symbols/sec.}$

The channel capacity in bits/sec is given by

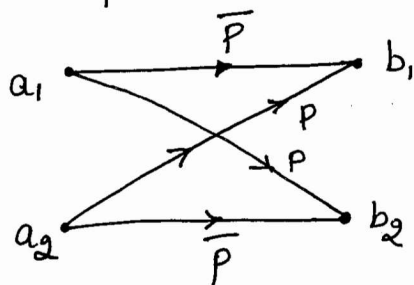
$$C = (\eta) (0.0817)$$

$$= (1000) (0.0817)$$

$$\boxed{C = 81.7 \text{ bits/sec.}}$$

④ Binary Symmetric Channel (BSC):

A Symmetric channel which has two inputs and two outputs, it is called a binary symmetric channel.



$$P(B/A) = \begin{matrix} & \begin{matrix} b_1 & b_2 \end{matrix} \\ \begin{matrix} a_1 \\ a_2 \end{matrix} & \begin{bmatrix} \bar{P} & P \\ P & \bar{P} \end{bmatrix} \end{matrix}$$

$$\bar{P} = 1 - P$$

The procedure to find $H(B/A)$ and C is same as in the symmetric channel,

$$\therefore \text{we have } c = \log_2 s - h \text{ bps}$$

$$\text{we know } h = \sum_{j=1}^s P_j \log_2 \frac{1}{P_j}$$

But in this case $s=2$, hence h is

$$h = \bar{P} \log_2 \frac{1}{\bar{P}} + P \log_2 \frac{1}{P}$$

$$\therefore c = \log_2 2 - \bar{P} \log_2 \frac{1}{\bar{P}} + P \log_2 \frac{1}{P}$$

$$c = 1 - \bar{P} \log_2 \left(\frac{1}{\bar{P}} \right) + P \log_2 \frac{1}{P}$$

$\therefore \boxed{c = 1 - h}$
This is the capacity of a binary symmetric channel.

Problems.

- ① A binary symmetric channel has the following noise matrix with source probabilities of $P(X_1) = 2/3$ and $P(X_2) = 1/3$.

$$P(Y/X) = \begin{matrix} & \begin{matrix} Y_1 \\ Y_2 \end{matrix} \\ \begin{matrix} X_1 \\ X_2 \end{matrix} & \begin{bmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{bmatrix} \end{matrix}$$

- ① Determine $H(X)$, $H(Y)$, $H(X,Y)$, $H(Y/X)$, $H(X/Y)$ & $I(X,Y)$
- ② find the channel capacity C
- ③ find channel efficiency and redundancy.

→ ① To find $H(X) = \sum_{i=1}^2 P(X_i) \log_2 \frac{1}{P(X_i)}$

$$H(X) = \frac{2}{3} \log_2 \frac{3}{2} + \frac{1}{3} \log_2 3$$

$$H(X) = 0.9183 \text{ bits/symbol.}$$

To find $H(Y)$

$$P(X, Y) = \begin{matrix} & Y_1 & Y_2 \\ X_1 & 1/2 & 1/6 \\ X_2 & 1/2 & 1/4 \end{matrix}$$

$$P(b_1) = 7/12 \quad P(b_2) = 5/12$$

$$H(Y) = - \sum_{j=1}^2 P(Y_j) \log_2 \frac{1}{P(Y_j)} = \frac{7}{12} \log_2 \frac{12}{7} + \frac{5}{12} \log_2 \frac{12}{5}$$

$$H(Y) = 0.9799 \text{ bits/symbol}$$

We know that $H(Y/X) = h = -\bar{p} \log_2 \frac{1}{\bar{p}} - p \log_2 \frac{1}{p}$

$$= -\frac{3}{4} \log_2 \frac{4}{3} - \frac{1}{4} \log_2 4$$

$$H(Y/X) = 0.8113 \text{ bits/symbol}$$

$$H(X, Y) = H(X) + H(Y/X)$$

$$= 0.9183 + 0.8113$$

$$H(X, Y) = 1.7296 \text{ bits/symbol}$$

$$H(X/Y) = H(X, Y) - H(Y)$$

$$= 1.7296 - 0.9799$$

$$H(X/Y) = 0.7497 \text{ bits/symbol}$$

To find $I(X, Y) = H(X) - H(X/Y)$ or $H(Y) - H(Y/X)$

$$= 0.9183 - 0.7497$$

$$I(X, Y) = 0.1686 \text{ bits/symbol}$$

ii) To find channel Capacity 'C'

$$C = 1 - h \Rightarrow 1 - H(Y/X) \Rightarrow 1 - 0.8113$$

$$C = 0.1887 \text{ bits/symbol}$$

iii) Channel efficiency $= \eta_{ch} = \frac{I(A, B)}{C}$

$$\Rightarrow \eta_{ch} = \frac{0.1686}{0.1887} \Rightarrow \eta_{ch} = 89.35\%$$

Channel Redundancy $= R_{ch} = 1 - \eta_{ch} =$

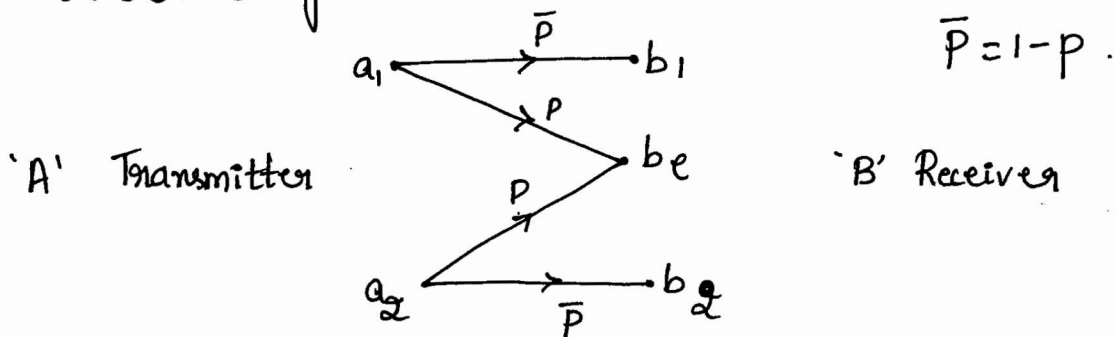
$$R_{ch} = 10.65\%$$

⑤ Binary Erasure Channel (BEC)

BEC is one of the most important channel used in Digital Communication.

If a symbol is received in error, it is just marked as error and discarded & a retransmission is requested from the transmitter using a reverse channel, till a correct symbol is received at the output. This ensures 100% correct data recovery. Since the error is totally erased in this channel, it is called "Binary Erasure Channel".

Channel Diagram.



ex:- So we have, $P(B/A) = \begin{matrix} & \begin{matrix} b_1 & b_e & b_2 \end{matrix} \\ \begin{matrix} a_1 \\ a_2 \end{matrix} & \begin{bmatrix} \bar{P} & P & 0 \\ 0 & P & \bar{P} \end{bmatrix} \end{matrix}$

In order to characterize the channel, find $H(A/B)$

We know $H(B/A) = h = \sum_{j=1}^S P_j \log_2 \frac{1}{P_j} \Rightarrow \boxed{\bar{P} \log \frac{1}{\bar{P}} + P \log \frac{1}{P}}$

if $P(a_1) = w$ then, $w + \bar{w} = 1$ &
 $P(a_2) = \bar{w}$ $P + \bar{P} = 1$

$\therefore$ we know $H(A) = \sum_{i=1}^2 P(a_i) \log_2 \frac{1}{P(a_i)}$

$\boxed{H(A) = w \log_2 \frac{1}{w} + \bar{w} \log_2 \frac{1}{\bar{w}}}$

$$P(A, B) = P(a_i, b_j) =$$

By xlying the first row
with w & second with $\bar{w}$

$$\begin{matrix} a_1 & \begin{matrix} b_1 & b_2 & b_3 \end{matrix} \\ a_2 & \begin{bmatrix} \bar{p}w & pw & 0 \\ 0 & p\bar{w} & \bar{p}\bar{w} \end{bmatrix} \end{matrix}$$

$$\therefore P(b_1) = \bar{p}w$$

$$P(b_2) = pw + p\bar{w} = p$$

$$P(b_3) = \bar{p}\bar{w}$$

$$\therefore P(A/B) = P(a_i/b_j) = \frac{P(a_i, b_j)}{P(b_j)} \quad \text{from JPM}$$

$$\therefore P(A/B) = \begin{bmatrix} \frac{\bar{p}w}{\bar{p}w} & \frac{pw}{p} & 0 \\ 0 & \frac{p\bar{w}}{p} & \frac{\bar{p}\bar{w}}{\bar{p}\bar{w}} \end{bmatrix}$$

$$\therefore P(A/B) = \begin{bmatrix} 1 & w & 0 \\ 0 & \bar{w} & 1 \end{bmatrix}$$

$$H(A/B) = \sum_{i=1}^n \sum_{j=1}^S P(a_i, b_j) \log_2 \frac{1}{P(a_i/b_j)}$$

$$= \cancel{\bar{p}w} \log 1 + pw \log \frac{1}{w} + p\bar{w} \log \frac{1}{\bar{w}} + \cancel{\bar{p}\bar{w}} \log 1$$

$$= pw \log \frac{1}{w} + p\bar{w} \log \frac{1}{\bar{w}}$$

$$H(A/B) = p \left[w \log \frac{1}{w} + \bar{w} \log \left(\frac{1}{\bar{w}} \right) \right]$$

$$\boxed{H(A/B) = p \cdot H(A)}$$

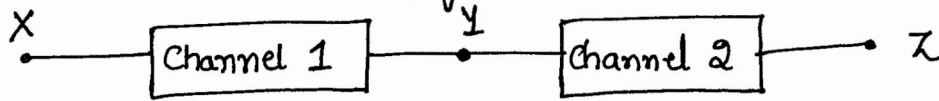
To find Capacity of the channel.

$$C = \text{Max} [I(A, B)] \eta_s$$

$$C = \text{Max} [H(A) - H(A/B)] \eta_s \Rightarrow C = [H(A) - p \cdot H(A)] \eta_s$$

$$C = \text{Max} [\bar{p} H(A)] \eta_s \Rightarrow C = \log_2 n \cdot \bar{p} \cdot \eta_s \Rightarrow \text{for } n=2, \eta_s=1 \Rightarrow 84 \text{ m/sec}$$

- ⑥ Cascaded channels: Two channels are connected in cascade as shown in below figure.



When the information is transmitted from X to Y through channel 1, there will be loss of information due to the noise in channel 1 & the mutual information ^{at the o/p} of channel 1 is

$$I(X, Y) = H(Y) - H(Y/X).$$

When the information is passed through Channel-II/2, there will be further loss of information & the mutual information at the output of the channel 2 would be

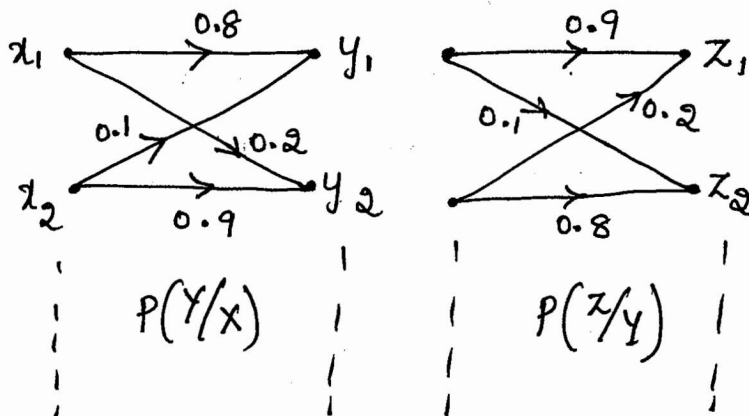
$$I(X, Z) = H(Z) - H(Z/X).$$

$$\therefore \boxed{I(X, Z) < I(X, Y)}$$

Problem.

- ① Two channels are Cascaded as shown in the figure, find the following.

$H(X), H(Y), H(Z), H(X, Z), H(Z/X)$ & $H(X/Z)$ for given probability of $P(X_1) = P(X_2) = 0.5$. Show that $I(X, Z) < I(X, Y)$



Solⁿ

we have $P(Y/X) = \begin{matrix} & y_1 & y_2 \\ x_1 & \begin{bmatrix} 0.8 & 0.2 \end{bmatrix} \\ x_2 & \begin{bmatrix} 0.1 & 0.9 \end{bmatrix} \end{matrix}$ & $P(Z/Y) = \begin{matrix} & z_1 & z_2 \\ y_1 & \begin{bmatrix} 0.9 & 0.1 \end{bmatrix} \\ y_2 & \begin{bmatrix} 0.2 & 0.8 \end{bmatrix} \end{matrix}$

$$P(Z/X) = \begin{matrix} & z_1 & z_2 \\ x_1 & \begin{bmatrix} 0.76 & 0.24 \end{bmatrix} \\ x_2 & \begin{bmatrix} 0.27 & 0.73 \end{bmatrix} \end{matrix}$$

$$\therefore P(X, Y) = \begin{matrix} & y_1 & y_2 \\ x_1 & \begin{bmatrix} 0.8 & 0.2 \end{bmatrix} \\ x_2 & \begin{bmatrix} 0.1 & 0.9 \end{bmatrix} \end{matrix} \times P(x_1) = 0.5$$

$$\times P(x_2) = 0.5$$

$$P(X, Y) = \begin{matrix} & y_1 & y_2 \\ x_1 & \begin{bmatrix} 0.4 & 0.1 \end{bmatrix} \\ x_2 & \begin{bmatrix} 0.05 & 0.45 \end{bmatrix} \end{matrix} \quad ||| \quad P(X, Z) = \begin{matrix} & z_1 & z_2 \\ x_1 & \begin{bmatrix} 0.38 & 0.12 \end{bmatrix} \\ x_2 & \begin{bmatrix} 0.135 & 0.365 \end{bmatrix} \end{matrix}$$

$$P(y_1) = 0.45 \quad P(y_2) = 0.55 \quad P(z_1) = 0.515 \quad P(z_2) = 0.485$$

$$\therefore H(X) = 0.5 \log_2 \frac{1}{0.5} + 0.5 \log_2 \frac{1}{0.5} = \boxed{1 \text{ bits/symbol}}$$

$$H(Y) = 0.45 \log_2 \frac{1}{0.45} + 0.55 \log_2 \frac{1}{0.5} = \boxed{0.9927 \text{ bits/symbol}}$$

$$H(Z) = 0.515 \log_2 \frac{1}{0.515} + 0.485 \log_2 \frac{1}{0.485} = \boxed{0.9993 \text{ bits/symbol}}$$

$$H(X, Z) = 0.38 \log_2 \frac{1}{0.38} + 0.12 \log_2 \frac{1}{0.12} + 0.135 \log_2 \frac{1}{0.135} + 0.365 \log_2 \frac{1}{0.365}$$

$$\boxed{H(X, Z) = 1.81825 \text{ bits/symbol}}$$

$$H(Z/X) = \sum_{i=1}^2 \sum_{j=1}^2 P(x_i, z_j) \log_2 \frac{1}{P(z_j/x_i)}$$

$$= 0.38 \log_2 \frac{1}{0.76} + 0.12 \log_2 \frac{1}{0.24} + 0.135 \log_2 \frac{1}{0.27} + 0.365 \log_2 \frac{1}{0.73}$$

$$\boxed{H(Z/X) = 0.8182 \text{ bits/symbol}}$$

$$H(X, Z) = H(X/Z) + H(Z)$$

$$H(X/Z) = H(X, Z) - H(Z) = 1.81825 - 0.9993$$

$$\boxed{H(X/Z) = 0.81895 \text{ bits/symbol}}$$

$$H(X, Y) = 0.4 \log_2 \frac{1}{0.4} + 0.1 \log_2 \frac{1}{0.1} + 0.05 \log_2 \frac{1}{0.05} + 0.45 \log_2 \frac{1}{0.45}$$

$$H(X, Y) = 1.595 \text{ bits/symbol.}$$

$$\therefore I(X, Y) = H(Y) - H(Y/X)$$

$$H(Y/X) = H(X, Y) - H(X)$$

$$= 1.595 - 1$$

$$H(Y/X) = 0.595.$$

$$\Rightarrow I(X, Y) = 0.9927 - 0.595$$

$$I(X, Y) = 0.3977.$$

$$\Rightarrow I(X, Z) = H(Z) - H(Z/X)$$

$$I(X, Z) = 0.9993 - 0.8182$$

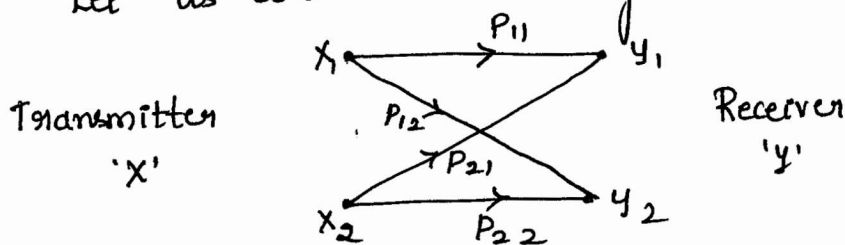
$$I(X, Z) = 0.1811.$$

$$\therefore I(X, Z) < I(X, Y)$$

$$\boxed{0.1811 < 0.3977}$$

Estimation of Channel Capacity By Muraga's Method.

Let us consider a binary channel as shown below



The channel matrix of the binary channel can be written as

$$P(Y_i/X_i) = \begin{matrix} & \begin{matrix} y_1 & y_2 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} \end{matrix}$$

To find the Channel Capacity of a binary channel, a method was suggested by Dr. S. Muroga, which is stated below:

Two Quantities θ_1 and θ_2 are found by using the matrix eqn

$$\begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} P_{11} \log P_{11} + P_{12} \log P_{12} \\ P_{21} \log P_{21} + P_{22} \log P_{22} \end{bmatrix}$$

expanding we get

$$P_{11} \theta_1 + P_{12} \theta_2 = P_{11} \log P_{11} + P_{12} \log P_{12}$$

$$P_{21} \theta_1 + P_{22} \theta_2 = P_{21} \log P_{21} + P_{22} \log P_{22}$$

The above eqn are simultaneous eqn which are solved to get θ_1 and θ_2 .

The channel Capacity 'C' of the binary channel is then calculated using the relationship

$$C = \log [2^{\theta_1} + 2^{\theta_2}] \text{ bits/symbol.}$$

IIIly for three Quantities

$$\begin{bmatrix} P_{11} & P_{12} & P_{13} \\ P_{21} & P_{22} & P_{23} \\ P_{31} & P_{32} & P_{33} \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix} = \begin{bmatrix} P_{11} \log P_{11} + P_{12} \log P_{12} + P_{13} \log P_{13} \\ P_{21} \log P_{21} + P_{22} \log P_{22} + P_{23} \log P_{23} \\ P_{31} \log P_{31} + P_{32} \log P_{32} + P_{33} \log P_{33} \end{bmatrix}$$

here θ_1, θ_2 & θ_3 are found

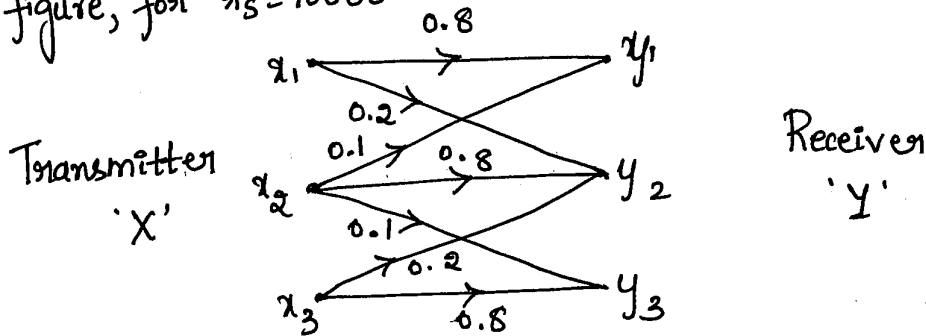
$$\therefore C = \log [2^{\theta_1} + 2^{\theta_2} + 2^{\theta_3}] \text{ bits/symbol.}$$

Generalizing the equation

$$\text{Channel Capacity 'C'} = \log [2^{\theta_1} + 2^{\theta_2} + \dots + 2^{\theta_m}] \text{ bits/symbol}$$

Problem.

- ① Find the Capacity of the discrete channel shown in figure, for $n_s = 10000$



→ the channel matrix can be constructed as

$$P(y_i/x_i) = \begin{matrix} & y_1 & y_2 & y_3 \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} & \begin{bmatrix} 0.8 & 0.2 & 0 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.2 & 0.8 \end{bmatrix} \end{matrix}$$

Using Muroga's method to find the channel capacity

$$\begin{bmatrix} 0.8 & 0.2 & 0 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.2 & 0.8 \end{bmatrix} \begin{bmatrix} Q_1 \\ Q_2 \\ Q_3 \end{bmatrix} = \begin{bmatrix} 0.8 \log 0.8 + 0.2 \log 0.2 \\ 0.8 \log 0.8 + 2 \times 0.1 \log 0.1 \\ 0.8 \log 0.8 + 0.2 \log 0.2 \end{bmatrix}$$

$$0.8 Q_1 + 0.2 Q_2 = -0.722 \quad \rightarrow \textcircled{1}$$

$$0.1 Q_1 + 0.8 Q_2 + 0.1 Q_3 = -0.922 \quad \rightarrow \textcircled{2}$$

$$0.2 Q_2 + 0.8 Q_3 = -0.722 \quad \rightarrow \textcircled{3}$$

Substituting $\textcircled{1}$, $\textcircled{2}$ & $\textcircled{3}$ eqn in calci we get

$$Q_1 = -0.6553 \quad Q_2 = -0.9887 \quad Q_3 = -0.6553$$

∴ The channel capacity is given by

$$C = \log [2^{Q_1} + 2^{Q_2} + 2^{Q_3}] = \log [2^{-0.6553} + 2^{-0.9887} + 2^{-0.6553}]$$

$$C = 0.8269 \text{ bits/symbol}$$

$$C = (0.8269) n_s$$

$$C = 8269 \text{ bits/sec}$$